

Respecting and Facilitating Children's Contributions to Research, Design and Evaluation in HCI

Course Submission

Janet C Read

University of Central Lancashire, Preston, UK

Matthew Horton

University of Central Lancashire, Preston, UK

ABSTRACT

Child Computer Interaction is concerned with the research, design, and evaluation of interactive technologies for children. Whilst many aspects of general HCI can be applied into this field, there are important adaptations to be made when conducting work for and with children throughout all stages of the design cycle. This course overviews the main tools and techniques in use by the CCI community against a backdrop of ensuring that children who work with our community feel valued and see the value in the work they contribute to, presented alongside examples and experiences from academia and industry. The course draws on the authors' experience within academia and industry hands on and provides useful checklists and tips to ensure children (and researchers and developers) get the most out of participation in HCI activities.

CCS CONCEPTS

• **Human-centered computing** → Interaction design; Interaction design process and methods.

KEYWORDS

Child Computer Interaction, Evaluation, Design, Research, Ethics, Children, UX research

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1 INTRODUCTION

Children are significant users of interactive technology and the recent lockdowns have forced them to spend significant chunks of time interacting with technologies that are often woefully suited to them [23]. Children are beginning to interact with technology at younger ages [12], and the value of the industry for interactive software for children is huge; but still there are many poorly designed products. In our own work we discovered first hand a plethora of difficulties that children had with mainstream technology [21].

There are also concerns about how much time children spend alone on technology rather than on playing with friends [1].

Designers and academics, therefore, need effective ways to better understand children and their contexts. When conducting research with, or designing technology for, children and families, researchers must take into consideration unique factors, that are less relevant for the general population. For example, research is needed to understand how children interact with technologies like robots [13], designers need to make products intended for learning more accessible to children who may have special needs [6], and children need to be empowered to give their opinions of, and their ideas towards, new innovations [2].

Engaging directly with children in research, design and evaluation can ensure that products are suited to their needs, but this engagement has consequences and the value of the engagement to children cannot be dismissed. The HCI community has an enthusiasm to listen to the voices of children through participatory practices [3] and through research [8], but in recent times the rights of the children, and their empowerment, have been a major theme [7], [9], [19]. UNICEF delivers a set of rights to children [26] in terms of their rights to be heard, protected and respected but it's not always easy to apply these rights in day to day situations as situations vary according to context – as an example, in one case it may be reasonable to limit a child's freedom to express themselves or have an opinion but in another instance that could be construed as oppressive [11]. As an example of a tension, in our own work, children aged 14 were prevented from being allowed to be photographed for a newspaper because their parents had decided they should not be photographed in school – the child's right to choose was taken from them.

Such tensions, and many others arise in our work with children in HCI. Spiel et al. [25] highlight a set of 'micro ethical' dilemmas that they encountered in their work with children, one example was the difficulty of walking away from children without a solution. This is tough when our rationale for working with children is often to improve their lives some way down the line. The extent to which children can consent, the difficulties around things like deception and the ad hoc selection of children for participation, were all flagged up as problematic for a set of students doing HCI research and design with children [5].

To help unpick these tensions, for the first time in a course on Child Computer Interaction, we bring children and their situations into the learning space by framing this new version of our course around a set of personas of children and a set of possible scenarios of participation. Whilst personas are widely used to design for children [29], [14], we use them in this course to help us design the 'research / design work' along the lines of a similar use in evaluation [10]. We will constantly 'talk to' our persona attendees during the course in order to focus on the value to the children of participation

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in whatever work we are planning / method we are explaining by stressing the need to enable, explain and empower participation.

Versions of this course have been delivered at previous CHI conferences and other ACM venues and these have separately covered research, evaluation, and design [24], [18], [20], [17]. This current version focusses on techniques that will ensure value to children as they participate in our research and design work. The course has been updated to include children and to consider how we might ensure that the children we interact with have the best possible time whilst the researchers get the best possible input, combining both academic research with children & industry research with children.

2 BENEFITS

This course will introduce attendees to core values, methods, and techniques for working with children in CCI and Children's UX with the focus being on the participation of children and the deliberate actions that can give them agency. Attendees will be given practical handouts and documentation including – personas, planning sheets, toolkits, and checklists. The course focuses on working with children in research, design and evaluation in ways that empower, and give value to, children.

At the end of the course, it is expected that attendees will:

- Better understand the unique characteristics of working with children in research, design, and evaluation, and be enabled and challenged to improve the experiences of the children they work with.
- Have a clearer understanding of the impact of method choices on children's agency and engagement and on research data, evaluation results and design ideas.
- Gain knowledge of techniques for the ethical recruitment of children (including the D4CR guide), the managing of studies and the writing and reporting of work.
- Be able to locate and use techniques that will be useful in general Child-Centered Design.
- Discover new methods and tools and feel empowered to apply them in their own work.

2.1 Intended Audience

This course is ideal for researchers, practitioners and designers who are interested in the research and design of digital environments for children (of all age groups). In past courses, participants included researchers, practitioners, and students.

2.2 Prerequisites

It is assumed that those attending have a basic understanding of HCI methods and/or UX research methods; outside of that there are no prerequisites.

3 CONTENTS

The course is structured in three parts, designed for delivery over three sessions. We have actively interspersed each session with three or more activities to ensure participant engagement in either online or face to face delivery.

Part 1: Children First.

We will begin the course with a discussion of how to put children first in our HCI work. This session will introduce a set of children who will come, as personas, to our course. From then on in, they will interrupt and challenge us as we go through the course ensuring that we focus on their needs. We will also introduce a set of scenarios that can happen when working with children which will be used to challenge some of our thinking as we go along. The theoretical part of this session will focus on children's rights and the ways that children can participate. We will use case studies and children's narratives to talk about when it might not be appropriate to work with children and on what measures we can take to gain value for children in participation.

Part 2: Methods for Working with Children.

This part of the course introduces a subset of methods for Research with Children, Design with Children and Evaluation with Children. We will introduce a set of tools and methods that have been shown to be effective when working with children. Research Methods include the PETT toolkit (in press) which can be used to describe the expertise of a sample of children. Design Methods that are described include Co-Design and Layered elaboration [28], and Obstructed Theatre [15] and we will stress the importance of using ideas from children in appropriate ways. Evaluation Methods include the MemoLine [27], This or That [30] and The Fun Toolkit [16] as well as discussion on planning evaluations and on carrying out observational and ethnographic work. During this session we will plan a session with the children we met in Part 1.

Part 3: Reporting and Ethics.

The last session is a discussion of reporting and ethics, which gives ideas for where we can go to learn more. We will discuss the CHECK toolkits for ethical work [22] as well as the Designing for Children's Rights (D4CR) guidelines [4]. Practical advice on the recruitment of children and on reporting both to the scientific community, and also to the children, is included in this section of the course.

3.1 Practical Work

A series of exercises is interspersed with the course content as seen in the attached PowerPoint file. Attendees carry out evaluations, plan a design session, and critique research. They additionally complete an ethical checklist and a D4CR evaluation.

3.2 Resources

The course materials will be placed in a shared Dropbox folder, and made available on the IDC website. Previous year's materials can be seen at <http://www.idc-sig.org/researchcourse>. Information about the instructors can be found on the ChiCI website (www.chici.org) and on Shuli Gilutz's website.

4 INSTRUCTOR BACKGROUND

Janet C Read is a Professor of Child Computer Interaction. Currently researching cross cultural CCI, childlike computing, the ethics of children's participation and visual programming, Prof. Read is the founding editor of the International Journal of Child Computer Interaction, a previous chair (on two occasions) of the ACM Interaction Design and Children (IDC) Conference and a former chair

of the ACM Child Computer Interaction Community. She is widely published in CCI and HCI.

Matthew Horton is a Senior Lecturer with over twenty years of experience in working with schools and industry in HCI and CCI. He has published on personas in CCI and in the active involvement of children in CCI work.

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